

Tyler Baker

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Mechanical engineer with hands-on experience in R&D, advanced manufacturing, prototyping, experimental testing, data acquisition systems, system integration, and project management.

Education

B.S. Mechanical Engineering, Engineering Management Minor

May 2024

Western Colo. Univ. - CU Boulder Engineering Partnership Program

GPA: 3.66

- Rady Full Tuition Merit Scholarship

Fall 2020 - Spring 2022

- HAAS Scholarship

Fall 2021 - Spring 2022

Relevant Coursework: Component Design, Thermodynamics I-II, Heat Transfer, System Dynamics, Manufacturing Processes, Product Development, Lean Six Sigma Manufacturing, Project Management

Work Experience

Research & Development Engineer

Aug. 2024 - Jan. 2026

Western Colorado University/Rocky Mountain Biological Laboratory (Gunnison, CO)

- Designed and deployed ground and UAV field-based sensor and mechanical systems for high-alpine environments, accounting for strict power, cost, and field deployment constraints
- Evaluated sensor system performance utilizing field data to inform iterative design revisions
- Contributed custom sensor and data collection systems in support of 3 scientific publications, 1 M.S. thesis, 1 PhD dissertation, and 2 hydrology projects

Mechanical Engineering Internship

May - Aug. 2023 & 2024

Blister Labs/Western Colorado University (Gunnison, CO)

- Utilized custom load and impact testing fixtures to evaluate compliance and damping characteristics of mountain bike wheels under controlled loading scenarios
- Integrated sensor and motion capture data to correlate experimental testing with real-world riding data and compare wheel designs and materials to inform rider wheel selection

Lead Makerspace Teaching Assistant

Jan. 2021 – May 2024

Western Colorado University (Gunnison, CO)

- Led daily makerspace operations including training TAs, supervising projects and shop equipment, and developing shop-specific SOPs and improvements

Projects

Adaptive Gravel Bike (Component Design Project)

Aug. 2022 – May 2023

- Collaborated with the Crested Butte Adaptive Sports Center and industry partners to design and build an ultralight bike frame utilizing SLS titanium additive manufacturing and bonded carbon fiber
- Applied failure mechanics principles and rider-specific load cases to inform structural design requirements while balancing strength-to-weight and user needs

Custom Steel Gravel Bike Frame (Independent Study)

Aug. 2023 – Dec. 2023

- Fabricated a custom fit gravel bike frame utilizing machining, TIG welding/sequencing, tube bending/notching, and fixturing techniques

SAE Baja Vehicle and Competition (Senior Capstone)

Aug. 2023 – May 2024

- Led as systems engineer to design, fabricate, and evaluate a custom off-road vehicle
- Contributed CAD modeling, manufacturing planning, fabrication, documentation, and project management under strict cost, time, and competition constraints

Technical Skills & Certifications

- Manufacturing: TIG/MIG Welding, Sheet Metal Fabrication, CNC & Manual Machining, Additive Manufacturing (FDM, SLA, SLS), Composite Bonding, Fixturing
- Software & Tools: SolidWorks (CSWA), Fusion 360, MATLAB, Python, Arduino IDE
- Process: Project Management (CPO), FE/EIT Certified, Lean/Six Sigma Principles, DFM/DFA, GD&T, FEA

Clubs/Organizations

- SAE Baja - Team Captain, ASME, WCU Mountain Bike Team